

Do materials matter? How surface representation affects presence in virtual environments

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ABSTRACT

This article reports the impact of different visual realism of materials applied to objects on perceived presence during an assembly task. The results of the experiment show that there is a significant difference between the more realistic scene and one where the surfaces of objects have been replaced with simpler, CAD-inspired visualizations. Despite these difference, both scenarios reach high values for presence and acceptance. Therefore, less detailed and less realistic rendering of surfaces might be sufficient to obtain a high presence and acceptance level in scenarios, which focus on manual tasks, if the associated drop in presence can be tolerated.

Keywords: Presence, User Studies, Virtual Reality, Material.

Index Terms: Human-centered computing → Visualization → Empirical studies in visualization; Human computer interaction (HCI) → HCI design and evaluation methods → User studies

1 INTRODUCTION

The introduction of virtual reality (VR) technologies has brought numerous benefits into many different branches of industry, for example enabling better and safer user training even in the early design stages of production. The goal of reaching high levels of presence in simulated tasks is often linked with the idea of ensuring the transferability of the virtually learned skills to the real tasks, as “many of the factors that appear to affect presence are known to enhance learning and performance” [1].

The term presence is described as a cognitive state – a “sense of being in the virtual environment” – that results from information processing of stimuli in the environment from various senses [2].

To achieve a high and fluent integration into the production process and to enable the above-mentioned advantages of VR with fast iteration cycles, creating virtual scenarios from existing CAD data should be as fast and easy as possible. Here, it must be noted that “CAD models emphasize the physical properties of solid objects” [3] and often “do not support techniques for modeling surfaces that are used by VR systems to significantly enhance the visual appearance of objects”. Thus the question is: Is it really necessary to change virtual materials and appearance of surfaces of CAD-created objects and their environments to reflect their real world behavior (with matching colors and shading fidelity) or is a CAD-like appearance of such objects sufficient enough to attain a high level of perceived presence when using VR applications.

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2 STATE OF THE ART

Multitudes of studies have researched different aspects of graphical fidelity and their influence on the users perceived presence. But often, these studies came to contradictory results: Zimmons et al. [4] showed that overall lighting and shadow quality as well as the usage of non-realistic surface textures does not always influence the experience of presence. Mania und Robinson [5] came to similar results: “the effect of the quality of rendering and in particular, shadow accuracy, employing the radiosity algorithm [...] showed [...] no effect of condition upon the presence and the lighting impressions datasets indicating that shadow accuracy did not affect participants’ sense of presence”.

In contrast, other studies, for example Hvass et al. [6] and Toczek [7], found that higher geometric realism combined with higher quality object surfaces lead to an increase in the feeling of presence, although it could not be determined which of these factors is the predominant one.

Overall, while this topic has been researched in a multitude of previous studies, there is still a noticeable lack of a clear consensus and universally valid statements of how surface representation affects performance.

3 EXPERIMENT

As described in the related work section, previous studies showed different results that could indicate that the additional expenses of creating realistic materials and surface visualizations could be avoided in VR-based training scenarios, while still attaining high levels of presence on the users’ side.

Therefore, we decided to compare a more realistic representation of the scene against one that used typical CAD color representation of objects. A between-subject study with the kind of object surface visualization as independent variable was conducted, avoiding the possible pitfalls of users realizing the nature of the study. In both experimental groups the participants wore an HTC Vive head mounted display and had to virtually assemble a simple toy truck. To study the influence of the surface properties of objects in a VR training environment, two different visualization concepts were developed: The first (hereafter called “realistic”) was crafted in a pseudo-realistic manner (see Figure 1). The surface albedo color and shading of the materials were designed to be good approximations of the real world counterparts. The second concept (hereafter called “CAD”) leaned heavily on the color design of common CAD software, with only simple shading with no highlights and randomly chosen colors picked from a predefined palette, like concurrent CAD software is using by default (see Figure 2).

Both experimental groups had to fulfil the same experimental procedure: after they read the participant information, they completed a quick tutorial to learn how to move and interact in the virtual environment and then started with the assembly task.



Figure 1: „Realistic” environment with objects possessing materials with color and surface shading close to the real world.



Figure 2: „CAD” environment with objects showing simply shaded materials with a CAD-inspired artificial and random color scheme

After finishing the assembly task, the participants filled out the questionnaires. Overall 66 participants – 34 in the “realistic” condition (22 female, 12 male, mean age: 24.38 (SD=3.61)) and 32 in the “CAD” condition (22 female, 10 male, mean age: 24.28 (SD=4.34)) participated in the experiment. All had normal or corrected-to-normal vision, participated in the study voluntarily and were allowed to abort the experiment at any time. The perceived presence was assessed with a shortened version of the ITC-SOPI [8], which included 12 items and was ranked on a five-point Likert scale. The ITC-SOPI contains four factors – sense of physical space, engagement, ecological validity and negative effects – which were measured by the three top loading items per scale.

4 RESULTS

For the evaluation of the influence of different surface representation, we compared the results of the presence factors of the ITC-SOPI between the different concepts. Figure 3 shows the means and standard deviations of the presence factors for both experimental groups. A Shapiro-Wilk test showed that the presence factors are not normally distributed. Therefore, the data was analyzed with a Mann-Whitney-U test at the 5 % significance level. The test showed no significant differences for the negative effects factor ($U = 487.00$, $Z = -0.81$, $p = 0.421$).

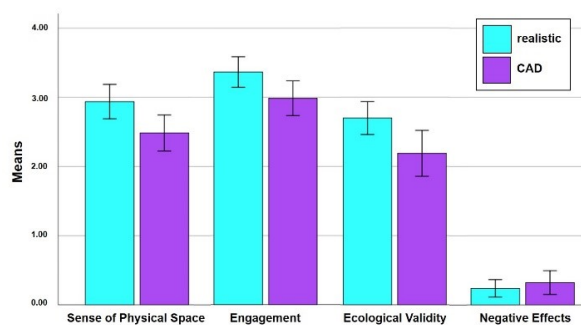


Figure 3: Means of the presence factors for both experimental groups

The test showed significant deviations concerning the presence factors sense of physical space ($U = 371.00$, $Z = -2.25$, $p < .001$), engagement ($U = 374.00$, $Z = -2.21$, $p < .001$) and ecological validity ($U = 390.00$, $Z = -1.99$, $p < .001$).

5 DISCUSSION AND CONCLUSION

The results show that the surface color and shading of objects significantly affects the presence and that a more natural look of the objects obtain higher presence values. Although the results of the CAD condition stand in strong contrast to the normal condition, it also reached overall high presence values. Thus, it can be concluded that sacrificing surface realism to decrease cost and effort of creating such VR training scenarios will lead to a lower observed presence for the user, but still might be feasible in certain scenarios, as the decrease is – while significant – not as strong as to completely diminish or inhibit the sense of presence.

Because the conducted study considered only two kinds of surface visualization and material simulation under a specific scenario focusing on manual tasks, the transferability of the results on other scenarios is limited. To proof the findings on other virtual scenes, a following study should address more variants and compare the surface visualization of interactive objects with different surface and material characteristics in the environments of these objects, as well as combinations of both.

ACKNOWLEDGMENTS & FUNDING

This project is co-financed with tax money based on the state budget, passed by the representatives of the Saxon Landtag.



Diese Maßnahme wird mitfinanziert durch Steuermittel auf der Grundlage des vom Sächsischen Landtag beschlossenen Haushaltes.

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